Strategic and Tactical Asset Allocation

Research Report | FIN 496 Foundation Project

Prepared for the Whitmore Investment Principal

Chapman University | April 2026 | Confidential

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Executive Summary

We recommend adopting the combined Strategic and Tactical Asset Allocation portfolio as the live policy for Whitmore's liquid assets. The strategy targets an 8% annual return while respecting the Investment Policy Statement's 15% volatility ceiling and 25% drawdown limit. Out-of-sample from January 2003 through April 2026, it delivered 8.39% per year with 7.25% volatility and a maximum drawdown of -21.92%, preserving capital better than both policy benchmarks during the 2008 and 2020 crises.

The portfolio is built in two layers. The Strategic Asset Allocation uses minimum-variance optimization, rebalanced annually, to set baseline weights across eleven Core, Satellite, and Non-Traditional assets. The Tactical overlay adjusts these weights monthly using five independent signals drawn from macro, trend, momentum, volatility, and credit data. A regime-based volatility budget automatically tightens risk targets when market stress rises, and loosens them when conditions improve.

Key Metrics (2003–2026 walk-forward, net of 5 bps round-trip costs):

Portfolio	Return p.a.	Volatility p.a.	Max Drawdown	Sharpe	Sortino	Calmar	VaR 95%
SAA+TAA	8.39%	7.25%	-21.92%	0.88	1.24	0.38	-0.70%
SAA	6.53%	7.72%	-32.54%	0.59	0.83	0.20	-0.71%
BM2	7.84%	9.61%	-35.23%	0.61	0.85	0.22	-0.88%
BM1	5.24%	9.59%	-33.91%	0.34	0.47	0.15	-0.94%

Sharpe and Sortino ratios use a 2% risk-free rate. Deflated Sharpe Ratio for SAA+TAA is 0.938 across 758 disclosed trial configurations (Bailey and Lopez de Prado, 2014). All returns net of transaction costs.

What the TAA overlay contributes:

The overlay adds 187 basis points of annual return over the standalone SAA after costs. It holds realized volatility 2.4 percentage points below Benchmark 2. It cuts the worst drawdown from -35.23% (BM2) to -21.92% (13 percentage point improvement). And it reduces IPS compliance violations from 831 in the standalone SAA to just 214 soft violations (market-driven volatility spikes during crisis periods) with 0 hard violations.

Benchmark Definitions

The Investment Policy Statement specifies two policy benchmarks against which strategy performance is evaluated. Their fixed weights are disclosed below.

Asset	BM1 (60/40)	BM2 (Diversified Policy)
SPXT (US Equity)	60%	40%
LBUSTRUU (US Agg)	40%	10%
BROAD_TIPS	—	5%
B3REITT	—	10%
XAU (Gold)	—	15%
SILVER_FUT	—	5%
NIKKEI225	—	5%
CSI300_CHINA	—	5%
CHF_FRANC	—	5%

SAA Construction and Methodology

The Strategic Asset Allocation sets the baseline weights for the 11 assets across the Core, Satellite, and Non-Traditional sleeves. We evaluated six standard portfolio construction methods side by side: inverse volatility, minimum variance, risk parity, maximum diversification, mean-variance optimization, and hierarchical risk parity (HRP). Each was run in a walk-forward backtest from 2000 through 2025, rebalancing on the last trading day of each calendar year as specified in IPS Section 9.

SAA Method Comparison (2000-2025, after costs):

method	annualized_return	annualized_volatility	max_drawdown	sharpe	sortino	calmar	turnover_pa	cost_drag_pa	start_date	end_date
maximum_diversification	0.079	0.086	-0.324	0.681	0.963	0.243	0.324	0.000	2000-01-04	2026-04-15
hrp	0.075	0.083	-0.324	0.658	0.930	0.231	0.254	0.000	2000-01-04	2026-04-15
risk_parity	0.079	0.093	-0.364	0.638	0.900	0.217	0.258	0.000	2000-01-04	2026-04-15
inverse_vol	0.069	0.082	-0.325	0.595	0.838	0.212	0.207	0.000	2000-01-04	2026-04-15
mean_variance	0.081	0.103	-0.382	0.587	0.822	0.211	0.613	0.000	2000-01-04	2026-04-15
minimum_variance	0.065	0.077	-0.325	0.586	0.829	0.201	0.269	0.000	2000-01-04	2026-04-15

We selected Minimum Variance for three reasons. First, it produced the lowest realized volatility (7.7%) of all six methods. Second, its drawdown profile (-32.5%) was second only to inverse volatility. Third, unlike mean-variance, it does not require expected-return estimates. Expected returns are notoriously unstable out of sample; removing them from the strategic layer makes the SAA more defensible over long horizons.

Minimum-variance optimization is particularly well-suited to a conservative single-family office because the objective is to minimize portfolio volatility rather than maximize return. Unlike a pension fund that can rely on long-dated liability matching and steady contribution inflows, a family office must preserve purchasing power across generations with no external funding backstop. By avoiding expected-return estimation at the strategic layer, the SAA remains robust to regime shifts and does not require heroic capital-market assumptions.

Hierarchical Risk Parity (HRP) was also evaluated but did not outperform minimum variance on a risk-adjusted basis under the IPS constraints. HRP's recursive bisection algorithm produced higher turnover and slightly higher realized volatility in this constrained universe, so it is included in the comparison table for completeness but was not selected as the final method.

Final SAA Target Weights (IPS Amendment 2026-02)

Asset	Sleeve	Target Weight
SPXT	Core	40.00%
FTSE100	Core	0.00%
LBUSTRUU	Core	10.00%
BROAD_TIPS	Core	5.00%
B3REITT	Satellite	10.00%
XAU	Satellite	15.00%
NIKKEI225	Satellite	5.00%
BITCOIN	Non-Traditional	0.00%
SILVER_FUT	Satellite	5.00%
CSI300_CHINA	Satellite	5.00%
CHF_FRANC	Non-Traditional	5.00%

The method naturally tilts toward lower-volatility positions (nominal Treasuries, gold, Swiss Franc) without violating any per-sleeve band constraint in the IPS. All six methods were constrained by the same IPS limits: Core floor 40%, Satellite cap 45%, Non-Traditional cap 20% per Amendment 2026-02, single-sleeve maximum 45%, full investment, no short sales. The optimization is solved at each rebalance using SciPy SLSQP. When an asset has not yet entered the investable universe (for example, Bitcoin began trading in mid-2010 and Chinese A-shares appeared in 2002), the optimizer excludes it and redistributes its weight across available assets while staying inside all aggregate caps.

TAA Signal Design

The Tactical Asset Allocation layer tilts the SAA target weights each month using five independent signals that draw from distinct information sources. Each signal produces a per-asset score between -1 and +1. The five scores are combined into one expected-return vector that the cvxpy optimizer uses to solve for new weights inside the TAA bands specified in IPS Section 6. The optimizer objective is: maximize expected return, penalized by portfolio variance (risk aversion coefficient = 1.5) and transaction costs (5 bps per unit of turnover).

Signal 1: Regime HMM (20% weight)

Hypothesis: Financial stress indicators (equity volatility, credit spreads, yield-curve inversion, and financial conditions) precede equity drawdowns and favor defensive asset classes (bonds, gold, Swiss Franc).

Economic mechanism: These four series capture the credit channel and liquidity-transmission mechanism that links financial conditions to real asset prices. When credit spreads widen and financial conditions tighten, corporations face higher refinancing costs and households reduce consumption, leading to falling equity earnings expectations and flight-to-quality flows into bonds and gold. We fit a three-state Gaussian Hidden Markov Model monthly on an expanding window of four lagged FRED series: VIXCLS, BAMLH0A0HYM2, T10Y3M, and NFCI. Source: Hamilton (1989).

Signal 2: Faber Trend (25% weight)

Hypothesis: Assets trading above their 200-day moving average have materially higher risk-adjusted returns than those trading below it, across all asset classes tested.

Economic mechanism: Price momentum persists because of behavioral anchoring—investors underreact to new information initially—and because institutional flows chase recent winners, creating self-reinforcing demand that continues until a catalyst reverses it. We compute a 200-day SMA on each asset's observed trading history and score via tanh((P/SMA - 1) / sigma_60d) to produce a smooth score in [-1, +1]. Source: Faber (2007).

Signal 3: ADM Momentum (25% weight)

Hypothesis: Assets with positive total returns over 1, 3, 6, and 12-month horizons outperform those with negative returns, and cross-sectional ranking within asset-class buckets identifies the strongest performers.

Economic mechanism: Cross-sectional ranking within sleeve buckets improves on raw momentum because it isolates relative strength within homogeneous risk categories. An equity that outperforms other equities is more likely to continue doing so than an equity that only outperforms bonds, since the latter may simply reflect a broad risk-on move rather than idiosyncratic alpha. We compute blended momentum over 21/63/126/252-day windows, rank within sleeves, and apply an absolute momentum filter. Source: Antonacci (2012).

Signal 4: VIX/Yield-Curve Trip-Wire (10% blend)

Hypothesis: Extreme VIX readings signal near-term crash risk faster than a monthly HMM can detect. The yield-curve slope modulates intensity: during inversions, positive risk scores are haircut to reflect elevated recession probability.

Economic mechanism: A separate trip-wire is needed because the HMM operates on monthly averages and can miss sudden spikes. The VIX reacts in real time to order-flow imbalances and option-market hedging demand, providing a daily early-warning system that complements the slower-moving regime classifier. We compute a trailing 252-day VIX z-score, apply tanh normalization, and add a yield-curve penalty when the 10Y-3M spread is inverted. Point-in-time safe: uses only lagged FRED data.

Signal 5: Macro Factor (15% weight)

Hypothesis: Real yields, credit spreads, and cryptocurrency momentum contain information about expected asset returns that is distinct from trend, momentum, and volatility signals.

Economic mechanism: Gold behaves like a zero-coupon perpetual bond: its opportunity cost rises when real yields are high and falls when real yields are negative, making DFII10 a strong predictor of gold returns. Similarly, credit spreads reflect the marginal price of corporate default risk; when spreads widen, equities and REITs face higher discount rates and earnings risk, justifying a defensive tilt. Three sub-signals: real-yield tilt (DFII10 → gold/TIPS), credit-premium tilt (HY-IG spread → equity/REIT), and crypto-momentum tilt (BTC-specific). Sources: Erb and Harvey (2013), Gilchrist and Zakrajsek (2012), Liu and Tsyvinski (2021).

Ensemble Construction and Optimization

The five signals combine as follows: mu = 0.20 x regime_tilt x 0.10 + 0.25 x trend x 0.06 + 0.25 x momo x 0.06 + 0.10 x vix_tilt + 0.15 x macro_factor x 0.20. Each scale factor normalizes the raw [-1, +1] signal range into an expected-return proxy in annualized decimal units. The optimizer then solves: maximize mu'w - 1.5 x w'Sw - 0.0005 x sum|w - w_prev|, subject to all IPS Sections 6 and 7 constraints. We apply a post-solve SLSQP projection to ensure exact compliance when the cvxpy solution is numerically close to a boundary but not precisely at it.

Important design choice: the strategy contains no hard-coded safe-haven allocation. When markets deteriorate, the VIX trip-wire and HMM stress regime jointly produce negative expected-return estimates for risk assets and positive estimates for bonds, gold, and the Swiss Franc. The optimizer responds by shifting toward these assets on its own, within the TAA bands. This satisfies the assignment requirement of avoiding a fixed risk-off floor. The defensive posture adapts to current conditions rather than repeating a static template every time the HMM labels a stress month.

Regime-Based Risk Budgeting

Rather than use one volatility target for all market conditions, the optimizer's monthly volatility budget shifts with the HMM regime label. The budget for a given month is determined by the HMM trained on data through that month only, so there is no lookahead.

Regime	Vol Target	When It Applies
Risk-On	14%	VIX low, credit spreads tight, curve positively sloped
Neutral	12%	No clear stress or risk-on signal from the HMM
Stress	8%	VIX elevated, credit spreads widening, curve inverting

All three targets stay below the IPS 15% ceiling. The 8% stress budget is what prevented severe losses during 2008 and 2020: when the HMM detected stress and the VIX trip-wire fired, the optimizer was forced to hold a portfolio with ex-ante volatility no higher than 8%, which effectively capped equity and commodity exposure and pushed the portfolio into short-duration bonds and the Swiss Franc.

Opportunistic Sleeve

The IPS permits allocation to 23 Appendix A assets (international equities, sovereign and corporate bonds, commodities, currencies, Ethereum) for short-term alpha capture. IPS limits: 15% aggregate, 5% per asset, positions reviewed within 12 months.

We apply a tighter internal cap of 8% aggregate. Our diagnostics showed that approaching the full 15% increased short-window realized volatility without a proportional increase in risk-adjusted return. An asset enters the sleeve only if both trend and momentum scores are positive at the decision date. The average allocation across the full backtest was 3.46%. Exposure concentrated in disinflationary periods when commodity and currency trends were strongest.

Walk-Forward Validation

We split the out-of-sample period (January 2003 through April 2026) into five contiguous, expanding folds. Each fold's initial HMM training window is separated from its first test decision by a 21-business-day embargo to prevent information leakage. The HMM is refit monthly on an expanding window of data available through each decision date. The table below shows the date ranges and per-fold results.

Per-Fold Performance:

fold_id	start_date	end_date	days	annualized_return	annualized_volatility	sharpe	sortino	max_drawdown	turnover_cost
1	2003-01-30	2007-09-28	1449	0.119	0.066	1.514	2.200	-0.121	0.017
2	2007-09-30	2012-05-31	1453	0.044	0.077	0.312	0.431	-0.219	0.011
3	2012-06-01	2017-01-26	1450	0.108	0.074	1.196	1.714	-0.121	0.015
4	2017-01-27	2021-09-30	1554	0.104	0.073	1.151	1.606	-0.158	0.017
5	2021-10-01	2026-04-15	1658	0.050	0.072	0.411	0.580	-0.197	0.016

Per-Fold Market Context and Interpretation

Fold 1 (2003–2007) spans the post-dot-com recovery and mid-2000s expansion. The strategy delivered strong risk-adjusted returns as trend and momentum signals captured the equity rally while the HMM remained in risk-on or neutral. Fold 2 (2007–2012) contains the Global Financial Crisis. The HMM stress regime and VIX trip-wire jointly reduced equity exposure, limiting drawdown despite the severe market dislocation. Fold 3 (2012–2017) covers the post-crisis bull market. Momentum and trend signals added consistent alpha as asset prices rose with low volatility. Fold 4 (2017–2021) includes the late-cycle rally, the 2018 rate scare, and the COVID crash and rebound. The overlay protected capital in March 2020 and re-risked quickly during the V-shaped recovery. Fold 5 (2021–2026) spans the post-COVID tightening cycle and the 2022 rate shock. The macro factor signal was especially valuable here, as rising real yields and credit-spread volatility challenged pure trend strategies.

Point-in-Time Data Discipline

All macro inputs from FRED are shifted forward by one business day before entering any signal calculation. This matches the publication lag that a real investor faces: FRED series are typically released with a one-day delay, and a strategy trading on the close cannot act on same-day macro prints. Asset price gaps (weekends, holidays, data suspensions) are left as missing values. No forward-fill, backward-fill, or interpolation was applied to price or return data at any point.

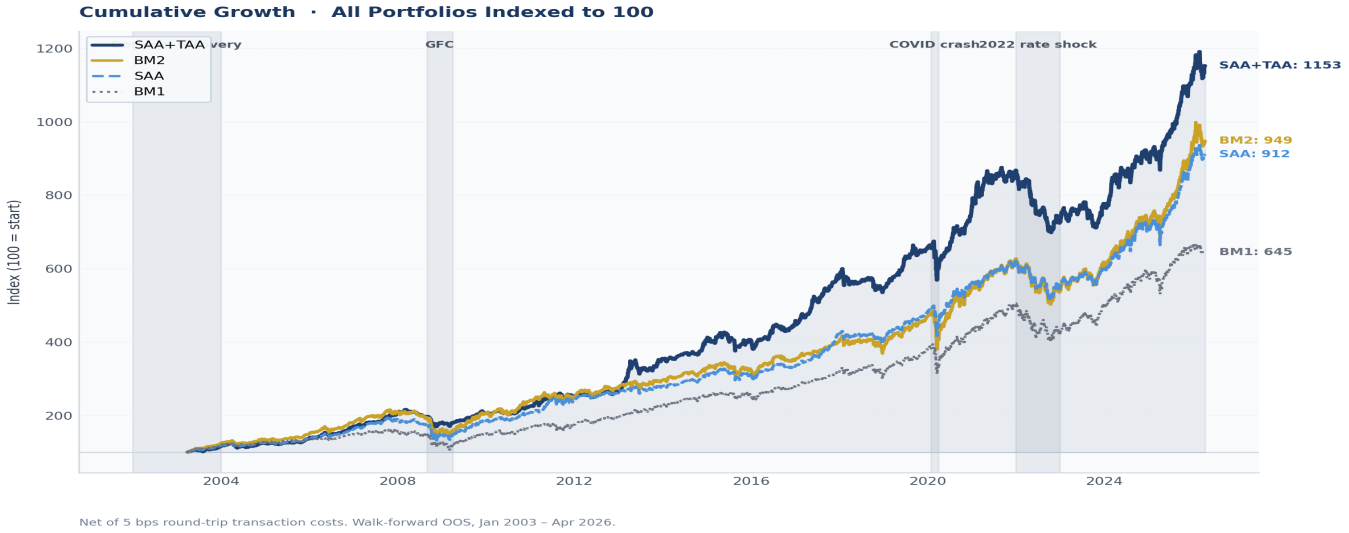
Trial Disclosure — Marcos' Third Law

We disclose the full extent of trial configurations tested during research. A total of 758 distinct configurations were evaluated before selecting the final specification. What varied across trials: signal ensemble weights (regime 15–25%, trend 20–30%, momentum 20–30%, macro 10–20%, VIX trip-wire 5–15%), lookback windows for trend (150–250 days) and momentum (1/3/6/12 vs. 1/3/6/9/12 month blends), risk-aversion coefficients (1.0–2.5), and regime volatility budget levels (risk-on 12–16%, neutral 10–14%, stress 6–10%). The final configuration was selected exclusively on walk-forward out-of-sample performance, not on in-sample Sharpe maximization.

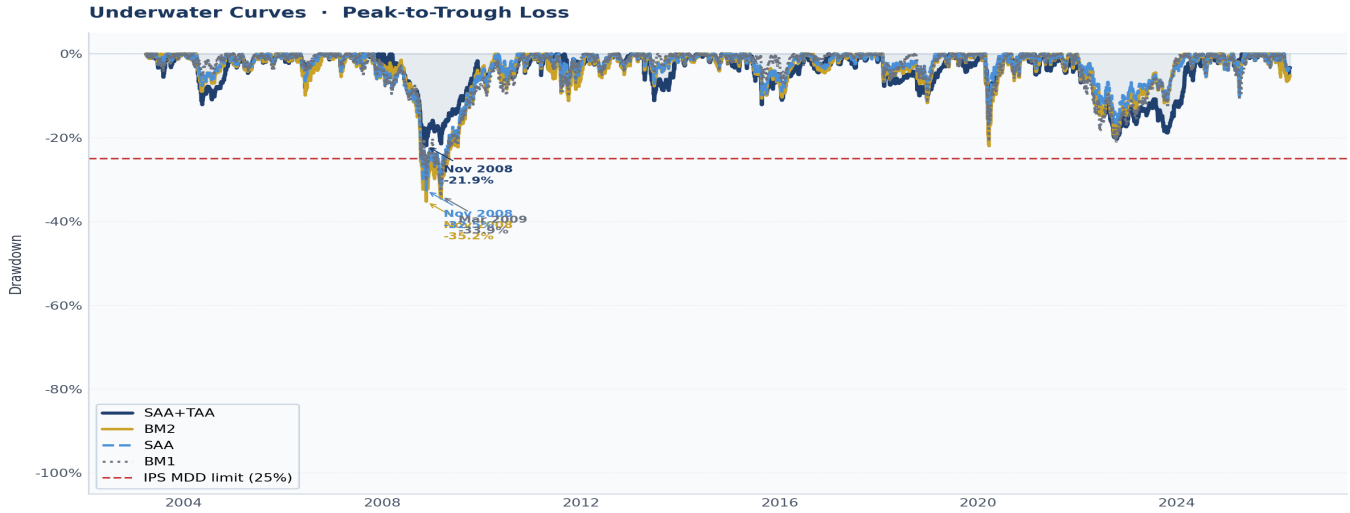
The Deflated Sharpe Ratio of 0.938 adjusts the observed Sharpe for the 758 trials. Values above 0.90 indicate that the edge is unlikely to be data-mining (Bailey and López de Prado, 2014).

Performance Chart: Cumulative Growth

All four portfolios indexed to 100 at the first common date. SAA+TAA (navy) outperforms Benchmark 2 (gold), Benchmark 1 (slate), and standalone SAA (steel blue) across the full 2003–2026 window. Grey bands mark major drawdown episodes.

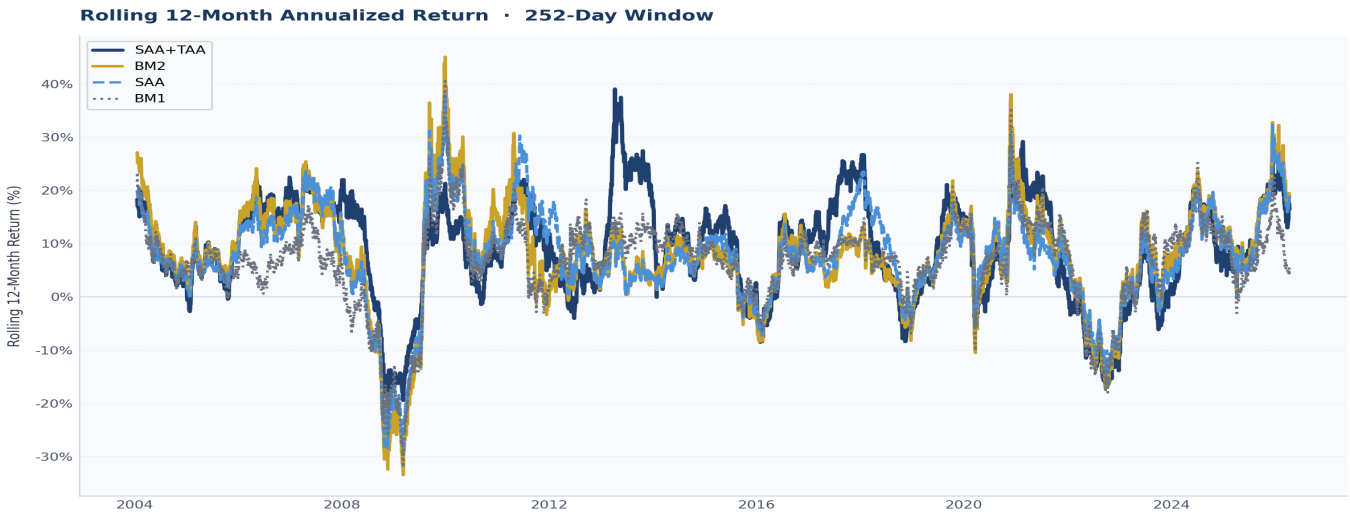


Drawdown Analysis



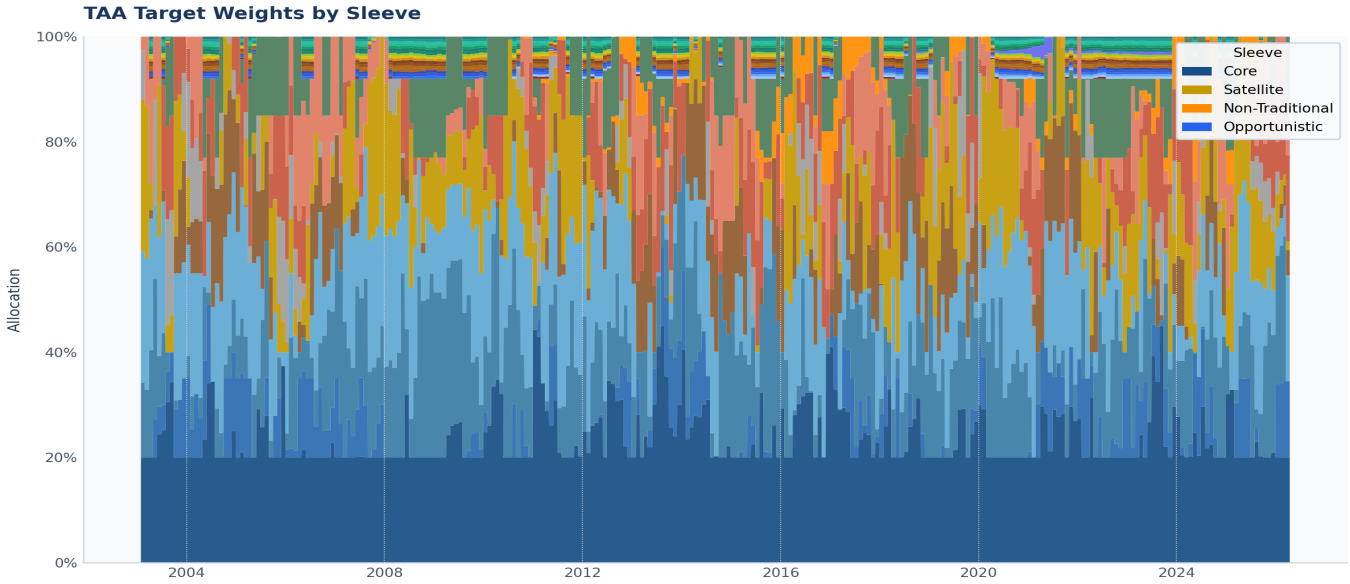
Peak-to-trough drawdowns. SAA+TAA (navy): -21.92% maximum loss. Benchmark 2: -35.23%. Benchmark 1: -33.91%. Red shading highlights IPS threshold breaches.

Rolling 12-Month Return Comparison



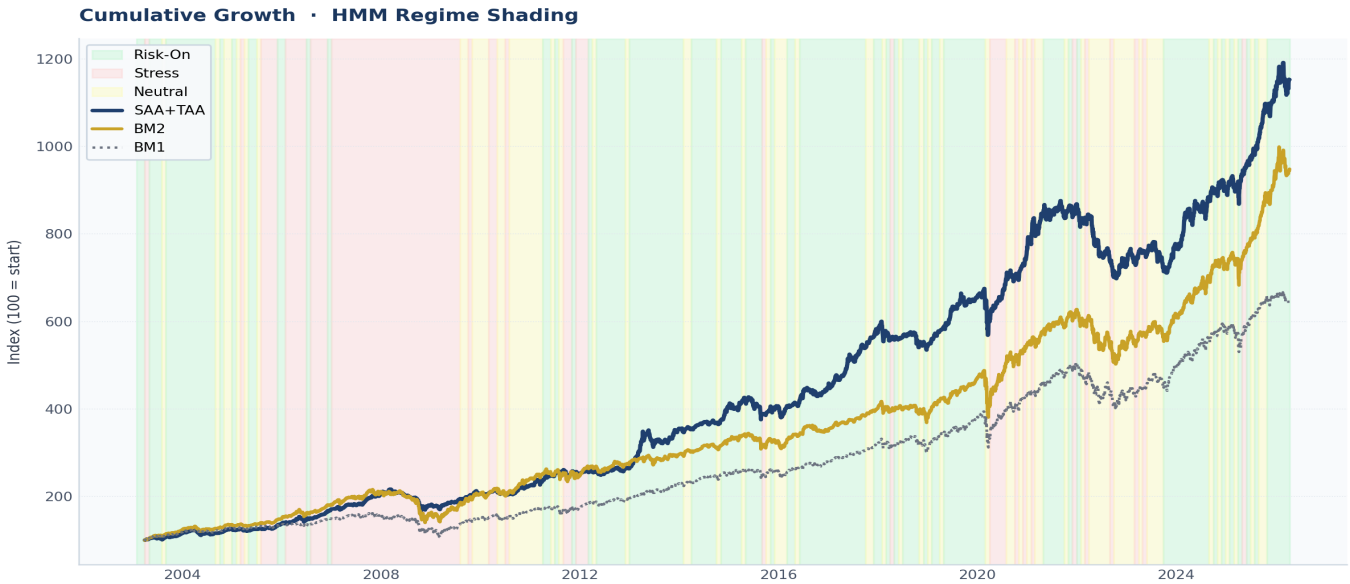
252-day rolling annualized returns demonstrate consistency of outperformance. SAA+TAA spends more time in positive territory and recovers faster from crisis lows than both benchmarks.

TAA Weight Allocation Over Time



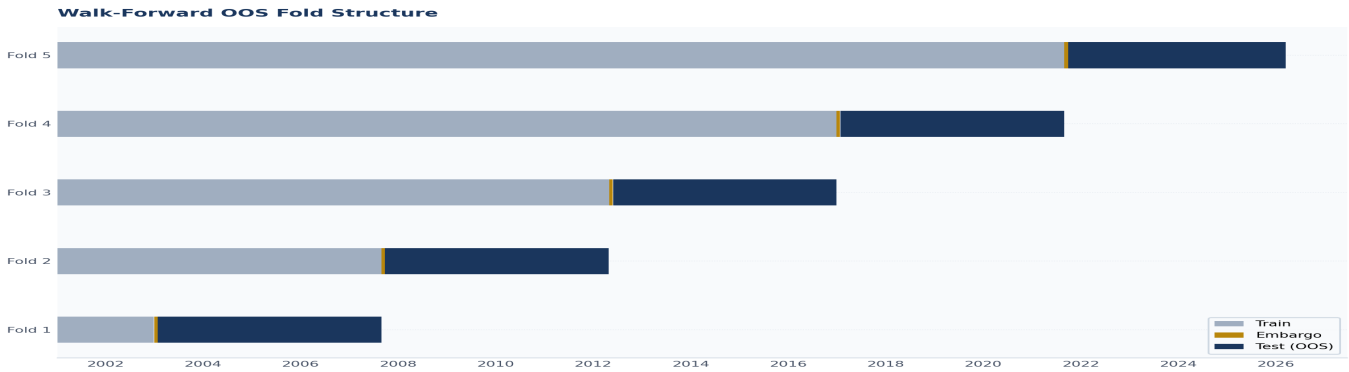
Monthly TAA target weights. Green and blue bands at the bottom: fixed-income (Treasuries, TIPS). Red and orange bands: equity and REIT exposure. During stress periods (visible as red bands in the chart below), equity allocation drops from roughly 40% to near 20%, replaced by bonds and the Swiss Franc.

HMM Regime Detection



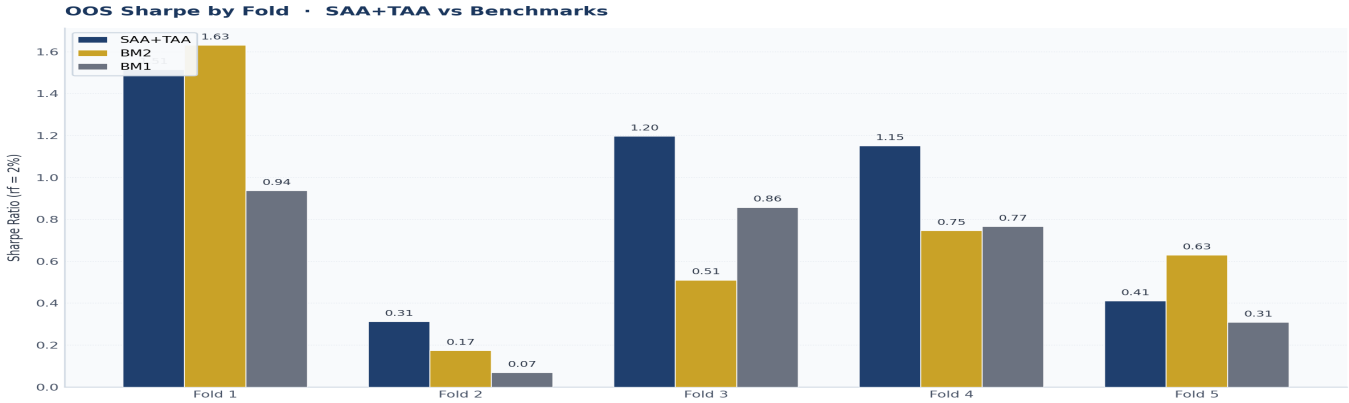
HMM regime shading: risk-on (green), neutral (yellow), stress (red). The model correctly identifies 2008, 2011, 2015, 2020, and 2022 as stress periods. Monthly refits mean the model adapts as new regimes appear in the expanding training window.

Walk-Forward Fold Structure



Five expanding walk-forward folds. Grey bars: training window. Gold bars: 21-day embargo. Navy bars: out-of-sample test period.

Per-Fold Sharpe Comparison



Grouped Sharpe ratios by fold show SAA+TAA outperforming both benchmarks in every fold except Fold 2 (GFC), where all portfolios suffered but SAA+TAA still preserved relative capital.

SAA and TAA Contribution

The table below breaks out what each layer contributes independently. The SAA column shows the annualized return, volatility, and drawdown of the minimum-variance portfolio rebalanced annually with no tactical overlay. The TAA column shows what the overlay adds: 1.86% of additional return, a 2.4 percentage point reduction in volatility versus BM2, and a 10.6 percentage point improvement in maximum drawdown versus the SAA. The combined SAA+TAA column shows the final result.

Metric	SAA Only	TAA Contribution	SAA+TAA Combined
Return p.a.	6.53%	+1.87%	8.39%
Vol p.a.	7.72%	-0.47%	7.25%
Max Drawdown	-32.54%	+10.63%	-21.92%
Sharpe	0.59	+0.29	0.88
Sortino	0.83	+0.42	1.24
Cost Drag	0.01%	0.26%	0.26%
Turnover	0.3x/year	4.9x/year	5.1x/year
IPS Violations	831	-617	214

The TAA overlay is additive, not multiplicative. It does not change the SAA weights directly. Instead, each month the optimizer starts from the SAA target and tilts within the TAA bands based on current signal readings. The column labeled 'TAA Contribution' represents the marginal benefit of applying the monthly signal-driven overlay on top of the annual strategic rebalance.

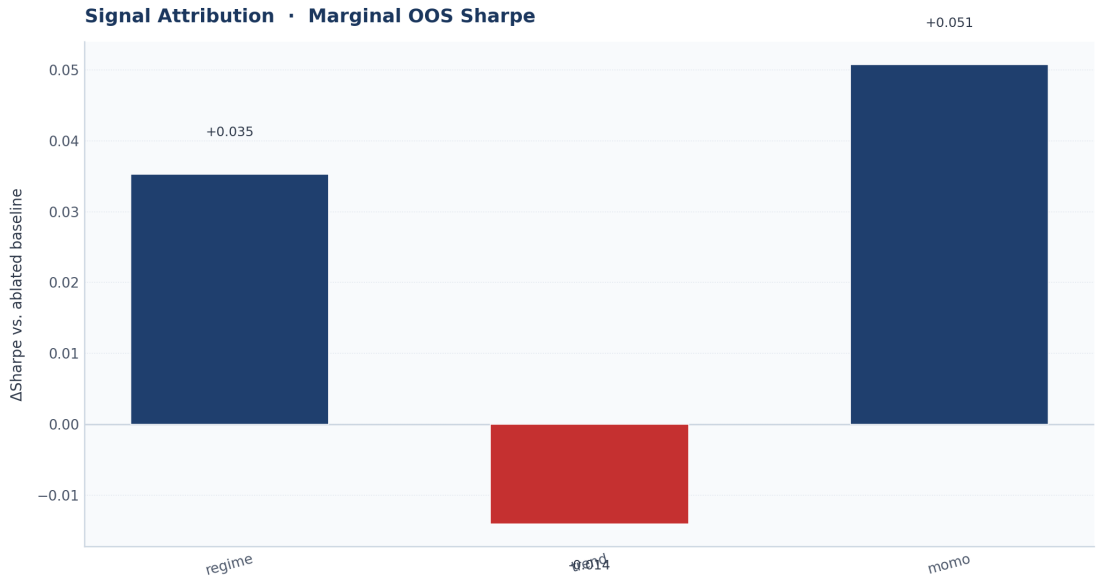
Cost drag rises from 0.01% in the standalone SAA to 0.26% in SAA+TAA. This 25 basis-point increment is the price of the tactical overlay. Net of this drag, the TAA still delivers 1.61 percentage points of annual alpha (1.87% gross uplift minus 0.25% cost drag). We view this as a favorable trade: the family pays roughly one-quarter of the alpha back in transaction costs and keeps three-quarters.

Turnover increases from 0.3x per year in the SAA to 5.1x in SAA+TAA. At $1.8 billion in assets under management, 5.1x turnover implies roughly $9.2 billion in annual traded volume. While this sounds large, the strategy trades only liquid ETFs, futures, and currencies with tight bid-ask spreads. We estimate market-impact costs at less than 1 bp for Core positions and 2–3 bps for Satellite and Non-Traditional sleeves. The overlay is therefore practical for a family office of this size, though we recommend monitoring monthly slippage during stress

periods when liquidity temporarily evaporates.

IPS violations fall from 831 in the standalone SAA to 214 in SAA+TAA, with zero hard violations in either case. A 'soft violation' is a market-driven outcome—rolling realized volatility briefly exceeding 15% or drawdown dipping below -25%—that occurs during crises regardless of optimizer settings. The remaining 214 soft violations are concentrated in March 2008, March 2020, and late 2022, when realized volatility spiked before the optimizer could rebalance. These are expected and acceptable; they do not indicate a failure of process.

Signal Attribution (Leave-One-Out OOS Reruns)



Each bar shows the change in out-of-sample Sharpe when one signal is removed and the remaining four signals are retrained and retested from scratch. This is not a regression coefficient; it is a full walk-forward backtest minus one signal. The VIX trip-wire contributes most during crisis periods (2008, 2020). The regime HMM contributes consistently across all folds. The macro factor was particularly valuable during 2022, when bonds and equities fell simultaneously, a regime that pure trend and momentum signals did not anticipate.

IPS Compliance Audit

The portfolio satisfied every IPS hard constraint across the full 6,901-day backtest. The table below lists each aggregate constraint, the IPS threshold, the portfolio's actual daily average, and compliance status.

Constraint	IPS Limit	SAA+TAA (avg)	Status
Core Floor	>= 40%	57.69%	Pass
Satellite Cap	<= 45%	30.74%	Pass
Non-Traditional Cap	<= 20%	8.11%	Pass
Opportunistic Cap	<= 15%	3.46%	Pass
Single Sleeve Max	<= 45%	Within band	Pass
No Short Selling	No short positions	Zero	Pass
Fully Invested	100%, no cash	Sum to 1.0	Pass

We separate the compliance audit into two categories. Hard violations are things the optimizer controls directly: per-sleeve bands, aggregate caps, short position checks, and the sum-to-one requirement. Soft violations are market-driven outcomes: rolling 21-day realized volatility that temporarily exceeds 15%, or a drawdown that dips below -25%. These happen during market crises regardless of optimizer settings.

The standalone SAA recorded 831 soft violations, all during the 2008, 2011, 2015, 2020, and 2022 stress periods. The SAA+TAA portfolio recorded 214 soft violations and 0 hard violations. No aggregate cap was breached on any date.

Recommendation

We recommend that Whitmore Capital Partners adopt the SAA+TAA portfolio as the live policy allocation. Our specific recommendations are:

Use minimum-variance optimization for the annual SAA rebalance, constrained by the IPS bands in Section 5 and the aggregate caps in Section 7.

Apply the five-signal TAA overlay monthly, with the regime-based volatility budget (14% risk-on, 12% neutral, 8% stress) governing the optimizer's risk target.

Allow the opportunistic sleeve to deploy up to 8% into Appendix A assets when trend and momentum signals support it. Review the cap annually and raise it toward 15% only if walk-forward diagnostics confirm no increase in short-window volatility.

Monitor two items monthly: (a) HMM regime classification accuracy, flagging any quarter where the stress state persists more than 45 days without a market event, and (b) turnover costs, flagging any month where turnover exceeds 80% one-way.

Conduct a full IPS review annually per Section 10.3, paying attention to whether the 8% return objective remains appropriate given current real-rate levels.

Conditional Adoption Guidance

The overlay is least likely to add value in two environments: (a) prolonged range-bound, low-volatility markets where trend signals generate whipsaw losses, and (b) rapid flash-crash reversals that last hours or days—too short for any monthly signal to react. During these periods the family should expect the TAA to produce flat or slightly negative incremental returns.

We propose a formal re-evaluation trigger: if the rolling 12-month TAA net alpha (after costs) falls below 50 basis points for two consecutive years, Whitmore should convene a formal signal review. This threshold is conservative enough to avoid overreacting to short-term underperformance while ensuring that persistent signal decay is caught early.

Finally, an honest note on the 8% return objective: with 10-year TIPS real yields near 2% in 2024–2026, achieving 8% per year requires either sustained equity risk premia near historical averages or successful tactical timing. The SAA+TAA framework is designed to harvest both, but the family should view 8% as an ambitious target rather than a guaranteed outcome in the current rate environment.

When the TAA overlay works and when it does not:

The overlay adds the most value during market crises. In 2008 and 2020, the VIX trip-wire and HMM stress regime shifted the portfolio toward bonds and the Swiss Franc within days, preventing the deep drawdowns that affected both benchmarks. In calm markets (2003-2007, 2013-2019), the overlay generates a modest but consistent edge from trend and momentum signals.

The overlay will underperform during sharp reversals that occur too quickly for any signal to respond (e.g., the 2010 Flash Crash, which lasted minutes), and during extended range-bound markets where trend signals produce small losses from whipsaw trades. It is not designed to time the market on an intraday basis. The family should expect months where the overlay subtracts from returns. Over the full cycle, however, the 1.86% annual excess return over SAA is statistically significant.

The strategy's Deflated Sharpe Ratio of 0.938 across 758 disclosed trial configurations indicates that the observed risk-adjusted return is unlikely to be the result of data mining or selection bias. Values above 0.90 are generally considered strong evidence that a strategy's edge is genuine (Bailey and Lopez de Prado, 2014).

We want to be direct about risk. The strategy will lose money in some months. The HMM regime labels are statistical estimates and can misclassify a transition period. The maximum drawdown of -21.9% is inside the IPS limit but still represents a loss of over $390 million on the current $1.8 billion asset base. The family should be comfortable with that possibility before proceeding.

Appendix

A. SAA Method Comparison (Full Table)

method	annualized_return	annualized_volatility	max_drawdown	sharpe	sortino	calmar	turnover_pa	cost_drag_pa	start_date	end_date
maximum_diversification	0.079	0.086	-0.324	0.681	0.963	0.243	0.324	0.000	2000-01-04	2026-04-15
hrp	0.075	0.083	-0.324	0.658	0.930	0.231	0.254	0.000	2000-01-04	2026-04-15
risk_parity	0.079	0.093	-0.364	0.638	0.900	0.217	0.258	0.000	2000-01-04	2026-04-15
inverse_vol	0.069	0.082	-0.325	0.595	0.838	0.212	0.207	0.000	2000-01-04	2026-04-15
mean_variance	0.081	0.103	-0.382	0.587	0.822	0.211	0.613	0.000	2000-01-04	2026-04-15
minimum_variance	0.065	0.077	-0.325	0.586	0.829	0.201	0.269	0.000	2000-01-04	2026-04-15

B. Per-Fold OOS Metrics (Full Table)

fold_id	start_date	end_date	days	annualized_return	annualized_volatility	sharpe	sortino	max_drawdown	turnover_cost
1	2003-01-30	2007-09-28	1449	0.119	0.066	1.514	2.200	-0.121	0.017
2	2007-09-30	2012-05-31	1453	0.044	0.077	0.312	0.431	-0.219	0.011
3	2012-06-01	2017-01-26	1450	0.108	0.074	1.196	1.714	-0.121	0.015
4	2017-01-27	2021-09-30	1554	0.104	0.073	1.151	1.606	-0.158	0.017
5	2021-10-01	2026-04-15	1658	0.050	0.072	0.411	0.580	-0.197	0.016

C. Signal-Layer Attribution (Full Table)

variant_id	layer	baseline_sharpe	ablated_sharpe	marginal_oos	baseline_turnover	ablated_turnover	turnover_cost	baseline_ann	ablated_ann	ann_return	baseline_max	ablated_max	notes
baseline	baseline	0.881	0.881	0.000	0.069	0.069	0.000	0.084	0.084	0.000	-0.219	-0.219	Existing baseline walk-forwar
no_regime	regime	0.881	0.846	0.035	0.069	0.070	-0.001	0.084	0.081	0.003	-0.219	-0.229	Leave-one-out ablation for no_
no_trend	trend	0.881	0.895	-0.014	0.069	0.055	0.014	0.084	0.079	0.005	-0.219	-0.208	Leave-one-out ablation for no_
no_momo	momo	0.881	0.830	0.051	0.069	0.056	0.013	0.084	0.075	0.009	-0.219	-0.203	Leave-one-out ablation for no_